A

Mini Project Report

on

GST Smart Manager

Submitted in partial fulfillment of the requirements for the

degree

Second Year Engineering – Computer Science Engineering (Data Science)

by

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Academic year: 2025-26

CERTIFICATE

This to certify that the Mini Project report on GST Smart Manager has been submitted by Yash Ballal (25207006) , Swarup Katke (25207009), Krishna Nikam (25207008) and Rohan Shelar (25207014) who are bonafide students of A. P. Shah Institute of Technology , Thane as a partial fulfillment of the requirement for the degree in Computer Science Engineering (Data Science) , during the academic year 2025-2026 in the satisfactory manner as per the curriculum laid down by University of Mumbai.

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1.

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Date:

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Chapter 1

Introduction

The Smart Tax Invoice Manager is a mobile application developed using Flutter and Dart to simplify business financial management. The application provides an efficient way to manage clients, generate invoices, calculate GST, and track payments.

In traditional systems, billing and tax calculations are often performed manually, which can lead to errors and inefficiency. This system automates these processes and provides a user-friendly interface to improve accuracy and productivity.

1.1. Purpose:

The purpose of the Smart Tax Invoice Manager is to develop an efficient and reliable system for managing business billing and financial operations in a simple and organized manner. The application helps users generate invoices, calculate GST automatically, and maintain proper records of transactions without manual effort.

In many small and medium businesses, billing and tax calculations are often done manually, which can lead to errors and poor record management. This system is designed to overcome these problems by providing automated GST calculation, accurate billing, and easy data management.

The application also aims to improve financial control by allowing users to track payments, manage clients, and view summaries of their business activities. Features like reminders and recurring invoices help users avoid missed payments and maintain consistency in billing.

Overall, the system focuses on reducing manual work, increasing accuracy, saving time, and providing a user-friendly solution for effective financial management.

1.2. Problem Statement :

Many small businesses still rely on manual billing methods or basic tools that lack proper GST calculation and financial tracking features.

This leads to:

Errors in tax calculation

- Difficulty in managing multiple clients and invoices
- Lack of proper financial tracking and reporting
- Missed payment deadlines due to no reminder system

Therefore, there is a need for a smart and automated system that can handle billing, GST calculation, and financial management in an efficient and user-friendly manner.

1.3. Objectives

- To automate GST calculations (CGST, SGST, IGST)
- To manage clients and invoices efficiently
- To generate financial reports and GST summaries
- To support multiple businesses within a single application.

1.4. Scope

The scope of the Smart Tax Invoice Manager includes providing a complete solution for managing business billing and financial operations through a mobile application. The system covers key functionalities such as client management, invoice generation, GST calculation (CGST, SGST, IGST), payment tracking, and financial reporting.

It allows users to handle multiple business profiles within a single application, making it suitable for individuals managing more than one business. The application also supports features like recurring invoice generation, payment reminders, and calendar-based tracking of important financial activities, which help in better organization and timely management.

The system is specifically designed for small and medium-scale businesses, shop owners, and freelancers who require a simple and efficient billing system. Its mobile-based nature ensures easy accessibility and usability anytime and anywhere.

However, the scope of the system is limited to standalone usage if cloud integration is not implemented. It may not support advanced enterprise-level features such as multi-user access, real-time synchronization across devices, or large-scale data handling.

Chapter 2

Proposed System

The proposed system, Smart Tax Invoice Manager, is a mobile-based application developed using Flutter and Dart to provide an efficient and automated solution for business billing and financial management.

Unlike traditional manual systems, the proposed system eliminates errors in invoice creation and GST calculation by automating the entire process. It offers a centralized platform where users can manage clients, generate invoices, track payments, and analyze financial data.

The system is designed with a user-friendly interface to ensure easy navigation and usability. It also supports multiple business profiles, allowing users to manage different businesses within a single application. Overall, the proposed system improves accuracy, saves time, and enhances productivity.

2.1. Features and Functionality

The Smart Tax Invoice Manager provides a wide range of features to support efficient billing and financial operations:

- Client Management

  Add, edit, and delete client details

  Maintain organized customer records

- Invoice Management easily

  Create and manage invoices

  Apply filters based on payment status (paid/unpaid)

- GST Calculation

Automatic calculation of:

CGST

SGST

IGST

Special GST billing mode for quick calculations

- Dashboard

  Displays overall business performance

  Shows recent transactions and summaries

- Multi-Business Support

  Manage multiple businesses in one app

  Easy switching between profiles

- Payment Reminders

  Set reminders for due payments

- Calendar Integration

  Track important financial dates and events

- Financial Reports

  Generate GST summaries and reports

  Analyze income and transactions

- Built-in Calculator

  GST calculation mode

Chapter 3

Project Outcomes

The final deliverables of the Smart Tax Invoice Manager project are categorized into functional outcomes and technical deliverables.

3.1 Functional Outcomes

1. User and Business Management
   - Users can create and manage multiple business profiles within the application.
   - Each business profile can store details such as business name, contact information, and GST-related data.
   - Users can easily switch between different business profiles.

2. Client Management Module
   - Users can add, edit, and delete client details.
   - Client information such as name, contact details, and transaction history is maintained.
   - The system ensures organized storage of customer records for easy access.

3. Invoice Management Module
   - Users can create invoices by selecting products/services and entering quantities.
   - The system automatically calculates totals and generates structured invoices.
   - Users can view, edit, and manage invoices with payment status tracking (paid/unpaid).

4. GST Calculation Module
   - The system automatically calculates GST components including CGST, SGST, and IGST.
   - A dedicated GST billing mode is provided for quick and accurate calculations.
   - Ensures compliance with tax requirements and reduces manual errors.

5. Payment Tracking and Reminder System
   - Users can track payment status for each invoice.

- Reminder notifications can be set for pending payments.
- Helps users maintain timely financial transactions.

6. Recurring Invoice and Calendar Module
   - Users can schedule recurring invoices for regular transactions.
   - Calendar integration helps track important financial dates and events.
   - Improves organization and time management.

7. Dashboard and Financial Reports
   - The system provides a dashboard displaying recent transactions and financial summaries.
   - Users can generate GST summaries and financial reports.
   - Helps in analyzing business performance effectively.

8. Built-in Calculator Module
   - Includes a standard calculator for general calculations.
   - Provides a GST-specific calculation mode for quick tax computations.

3.2Technical Deliverables

- Flutter Mobile Application – Developed using Flutter, providing a responsive and user-friendly interface for Android devices.

- Backend and Database (Firebase) – Implemented using Firebase for real-time data storage, authentication (if used), and secure data management.

- Source Code Management – Maintained using GitHub for version control and project tracking.

- Application Logic – Developed using Dart to handle GST calculations, invoice processing, and data operations.

- Documentation – Includes complete project report, user guide, and system description for proper understanding and usage.

3.3Measurable Success Criteria

- The application successfully generates invoices with accurate GST calculations.
- All core features such as client management, invoice generation, payment tracking, and reporting are implemented and tested.
- The system provides quick response time for data entry, retrieval, and calculations.
- Users are able to manage multiple businesses efficiently within a single application.
- The application ensures ease of use with a user-friendly interface and smooth navigation.
- Financial reports and summaries are generated correctly for better decision-making.

Chapter 4

Software Requirements

This section describes the software tools, technologies, and platforms used for the development, testing, and execution of the Smart Tax Invoice Manager application.

4.1 Development Tools

Tool / Technology — Purpose

- Flutter SDK — Used for developing the mobile application with a user-friendly interface
- Dart — Programming language used to implement application logic and GST calculations
- VS Code — Used as IDE for coding, debugging, and testing
- Firebase Console — Used for backend services such as database and data management
- Git — Used for version control to track changes in the project
- GitHub — Used for storing and managing the project repository

4.2 Frontend Dependencies (Flutter Packages)

Package — Purpose

- firebase_core — Used to connect the Flutter app with Firebase
- cloud_firestore — Used as a database to store client, invoice, and business data

- firebase_auth (if used) — Used for user authentication
- provider — Used for state management in the application
- image_picker — Allows user to upload images (like profile or business logo)
- intl — Used for formatting dates, currency, and invoice details

4.3 Hardware Requirements (for testing)

- Android device (minimum API level 21) for running the application
- Android Emulator for testing during development
- Stable internet connection for Firebase database access

4.4 Non-Functional Requirements

- Performance: The application should load quickly and provide smooth navigation between screens.
- Security: User data such as client details and invoices should be stored securely in Firebase.
- Usability: The interface should be simple and easy to use for all types of users.
- Reliability: The system should work properly without crashes and handle data efficiently.

Chapter 5

Project Design

The system architecture of the Smart Tax Invoice Manager follows a mobile-based structure where the Flutter application acts as the client and Firebase is used as the backend. The main components of the system include the Flutter mobile application, Firebase services, and internal modules such as invoice management, GST calculation, and reporting.

5.1 Block Diagram

The block diagram below represents the interaction between the main modules of the system.

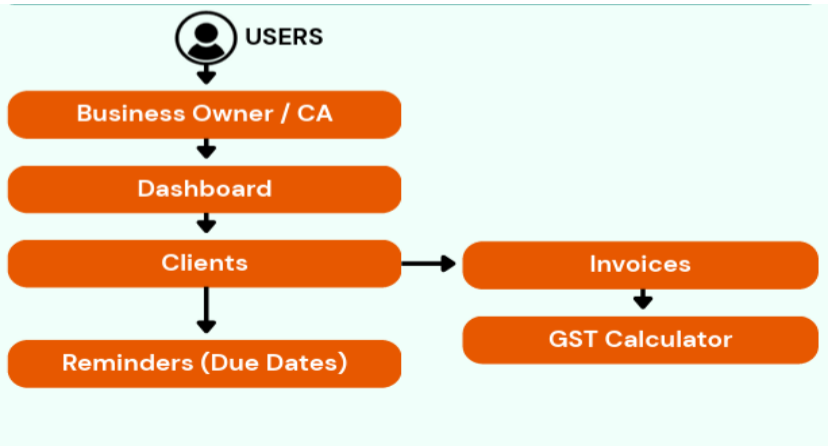


Figure 5.1: Block Diagram of Smart Tax Invoice Manager

- Users (Business Owner / CA)
  - The system is used by the business owner or accountant.
  - They perform all operations such as managing clients, invoices, and financial data.
- Dashboard
  - The dashboard provides an overview of business activities.

- It displays important information like total invoices, pending payments, and recent transactions.

- Clients Module
  - This module allows the user to add, update, and manage client details.
  - All customer information is stored and used for invoice generation.

- Invoices Module
  - The invoices module is responsible for creating and managing invoices.
  - It takes input from client data and generates structured bills.

- GST Calculator
  - This module automatically calculates GST (CGST, SGST, IGST).
  - It is connected to the invoice module to compute the final bill amount.

- Reminders (Due Dates)
  - This module helps track pending payments.
  - It notifies the user about due dates and unpaid invoices.

- Flow (Based on Diagram)
  - User → Dashboard → Clients
  - Clients → Invoices → GST Calculator
  - Clients → Reminders (Due Dates)

5.2 System Overview

The system works as a complete billing solution for managing business financial activities.

The user starts by creating a business profile and adding client details. After that, invoices can be generated by entering product or service information along with quantity and price. The system automatically calculates GST and provides the final bill amount.

All the data such as client details, invoices, and transaction records are stored in Firebase, which ensures proper data management and easy retrieval. The dashboard provides an overview of financial activities, including recent transactions and summaries.

The application also includes features such as payment tracking, reminders, and recurring invoices, which help users manage their billing cycle efficiently. The built-in calculator allows quick calculations, including GST-based operations.

Overall, the system provides a simple, organized, and efficient way to handle billing and financial management using a mobile application.

5.3 WorkFlow Diagram:

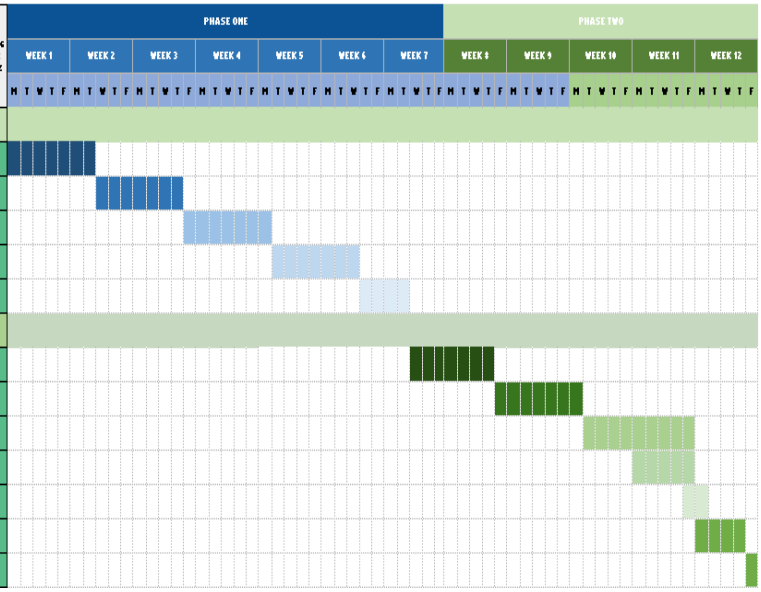
Figure 5.3: Workflow Diagram

Chapter 6

Project Scheduling

The project is planned over Three months. The Gantt chart below outlines the key tasks and their estimated timelines.

WBS NUMBER	TASK TITLE	TASK OWNER	START DATE	DUE DATE	DURATION(WEEKS)	PERCENTAGE OF TASK COMPLETE
1	**Project Conception and Initiation**					
1.1	Project setup & system architecture	A,B,C,D	1-10-26	1-16-26	1	100%
1.2	Firebase configuration & database design	A,B,C,D	1-17-26	1-23-26	1	100%
1.3	Authentication & user profile setup	A,B,C,D	1-24-26	1-30-26	1	100%
1.4	Profile enhancement	A,B,C,D	1-31-26	2-6-26	1	100%
1.5	Presentation I	A,B,C,D	2-7-26	2-10-26	1	100%
2	**Project Design and Implementation**					
2.1	Business setup & profile management	A,B,C,D	2-11-26	2-17-26	1	100%
2.2	Client management & invoice tracking	A,B,C,D	2-18-26	2-24-26	1	100%
2.3	Invoice generating & GST calculation	A,B,C,D	2-25-26	3-5-26	1	100%
2.5	Security implementation & performance optimization	A,B,C,D	3-1-26	3-5-26	1	100%
2.6	Testing & bug fixing	A,B,C,D	3-16-26	3-25-26	1	100%
2.7	Documentation	A,B,C,D	3-26-26	3-5-26	1	100%
2.8	Presentation II	A,B,C,D	4-6-26	4-10-26	1	100%



Chapter 7

Results

The Smart Tax Invoice Manager application has been successfully developed and tested based on the defined objectives. All main modules such as client management, invoice generation, GST calculation, and reporting are working properly and are well integrated within the system.

7.1 Implemented Features

- Business Management Module – Allows users to create and manage multiple business profiles with required details.
- Client Module – Users can add, update, and delete client information, and maintain proper customer records.
- Invoice Module – Users can create invoices by entering product/service details, quantity, and price. The system automatically generates the final bill.
- GST Calculation Module – Automatically calculates CGST, SGST, and IGST, ensuring accurate tax computation.
- Payment Tracking – Helps users track payment status (paid/unpaid) and manage pending transactions.
- Reminder and Recurring Features – Users can set reminders for pending payments and create recurring invoices for regular customers.
- Dashboard and Reports – Provides financial summaries, recent transactions, and GST reports for analysis.
- Calculator Feature – Includes both basic calculator and GST calculation mode for quick computations.

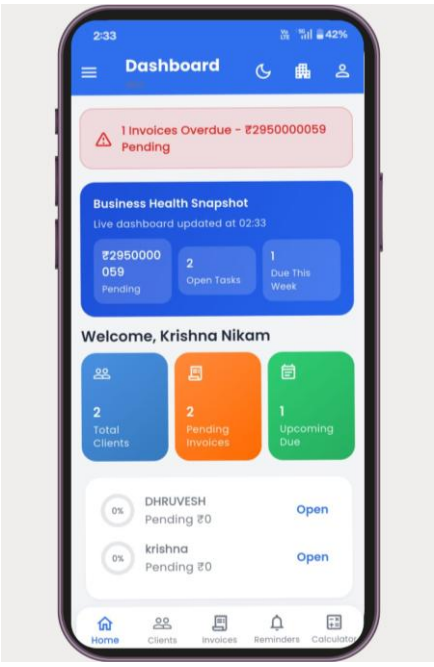
7.2 System Performance

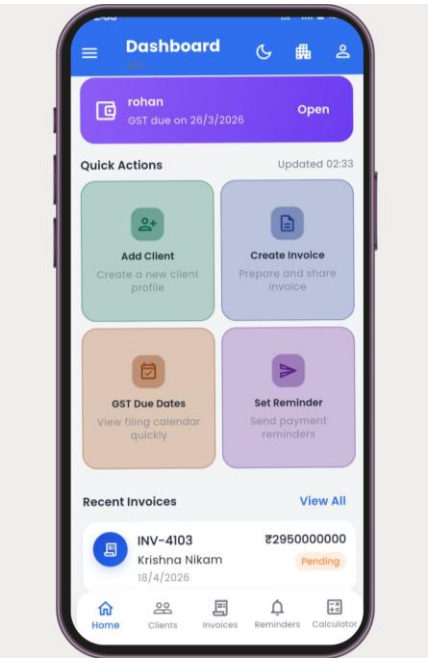
- The application runs smoothly on Android devices with good response time.
- Firebase ensures fast data storage and retrieval for client and invoice information.
- GST calculations are performed instantly and accurately.
- The system provides a simple and user-friendly interface, making it easy to use for different users.
- The application handles multiple business profiles efficiently without performance issues.

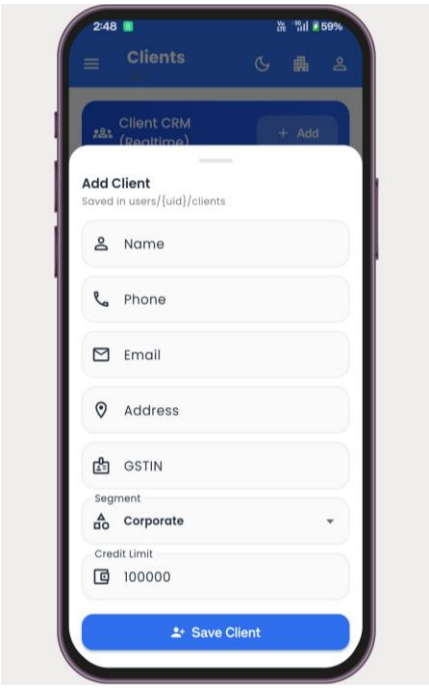
7.3 Final Deliverables

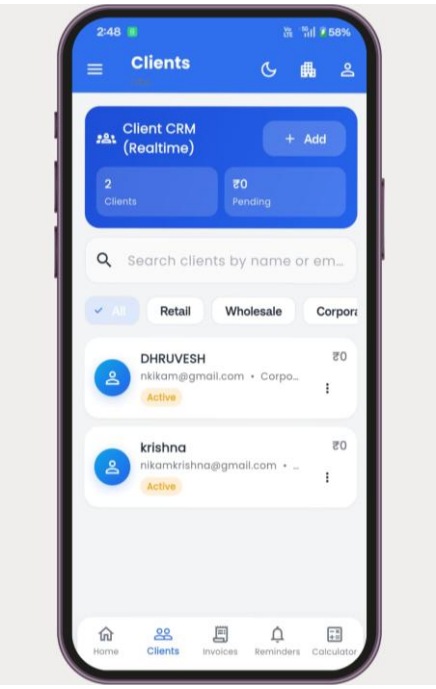
- Complete Flutter application source code stored on GitHub.
- Firebase backend with properly managed database for storing application data.
- Fully functional mobile application with all core features implemented.
- Project documentation including report and system description.

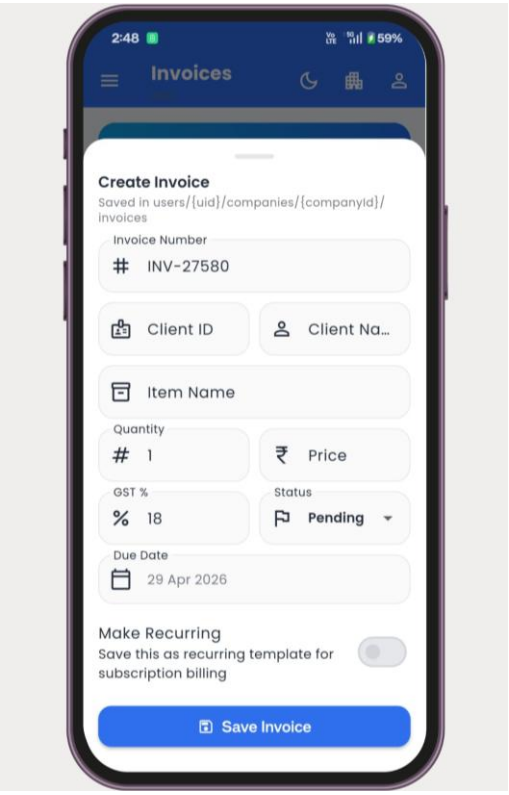
7.4 Visual Representation of Application







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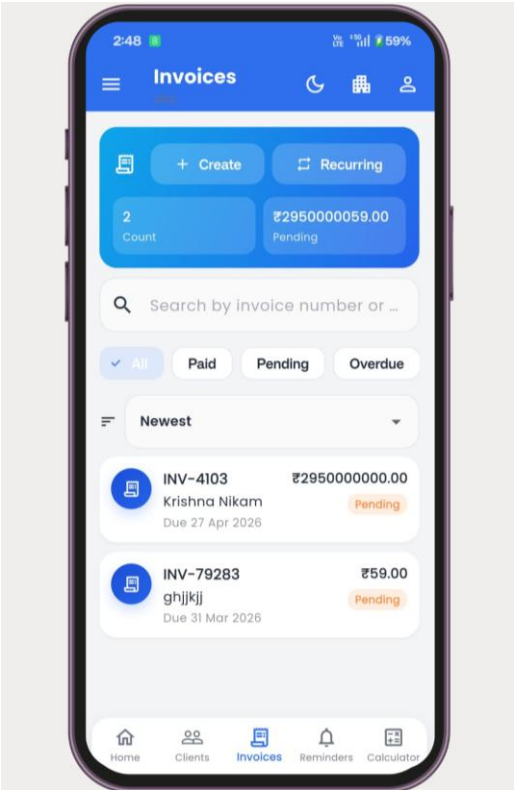
Invoices
Create Invoice
Saved in users/{uid}/companies/{companyId}/invoices
Invoice Number
INV-27580
Client ID
Client Na...
Item Name
Quantity
1
Price
GST %
18
Status
Pending
Due Date
29 Apr 2026
Make Recurring
Save this as recurring template for subscription billing
Save Invoice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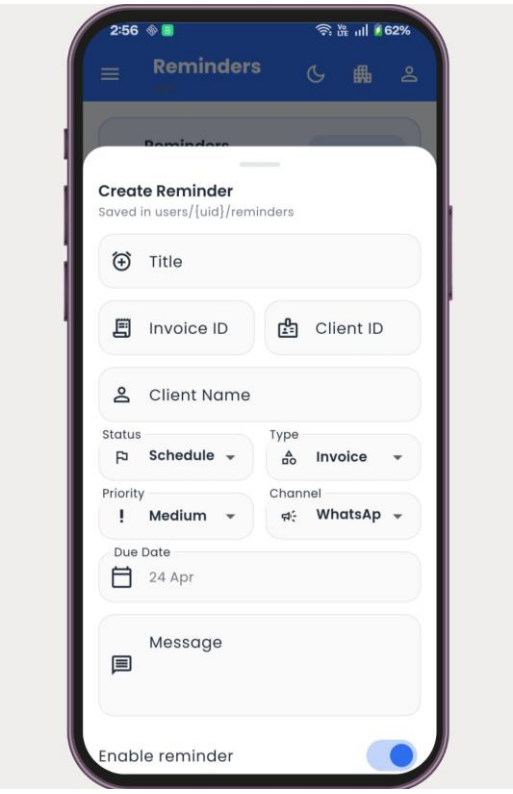
Reminders
Calculator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Reminders
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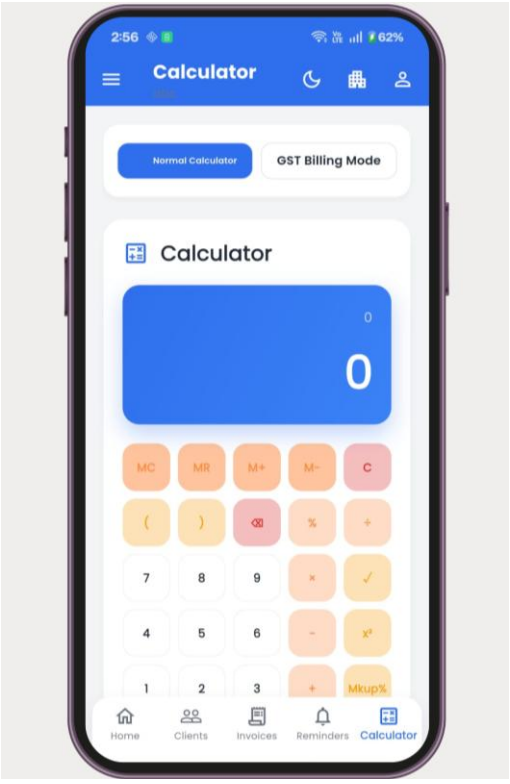
Due Date
24 Apr
Message
Enable reminder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Calculator
Normal Calculator
GST Billing Mode
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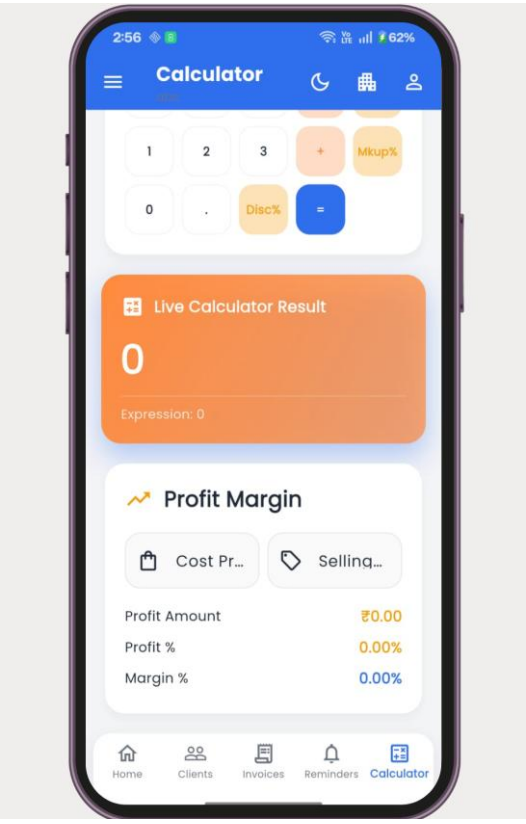
Calculator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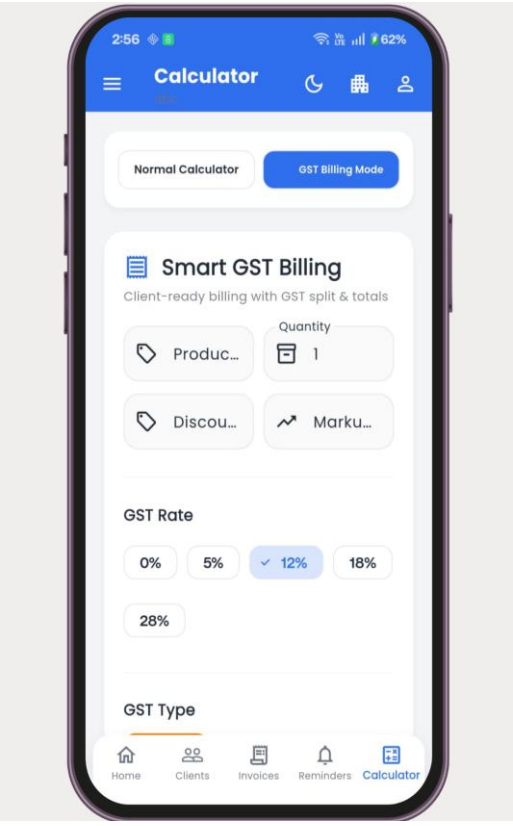
Calculator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Calculator
Normal Calculator
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Smart GST Billing
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Discou...
Marku...
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12%
18%
28%
GST Type
Home
Clients
Invoices
Reminders
Calculator

Chapter 8

Conclusion:

The Smart Tax Invoice Manager successfully addresses the challenges of manual billing and financial management by providing a centralized mobile application for invoice generation, GST calculation, client management, and payment tracking. The system simplifies business operations by automating tax calculations and maintaining organized financial records, which helps in reducing errors and saving time.

The use of Flutter and Firebase ensures smooth performance, real-time data handling, and easy access to information. The application provides a simple and user-friendly interface, making it suitable for small and medium-scale businesses.

The system also improves financial tracking by offering features such as reminders, recurring invoices, and detailed reports, allowing users to manage their business activities more efficiently.

Future enhancements can include features such as cloud backup, PDF invoice generation, online payment integration, and multi-user access for better scalability and usability.

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